Demo Client · DEMO-0010
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Field	Value
Document	Methodology of the data-quality and reconciliation review
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1. Purpose

This document records how the data was measured and reconciled, step by step, so that an auditor can repeat it. The review is not an audit and gives no assurance on the financial statements.

2. Inputs

Data set	Size	Fingerprint (SHA-256)
customers	840 rows	`1cda64a6c8808d34`
invoices	5,212 rows	`7fdf1307089369a6`
ledger	7,378 rows	`520f9294fe16a6d2`
bank	6,940 rows	`405588f5b1cb031d`
bank_balances	772 rows	`b98ce8755b81d1fe`
accounts	4 rows	`f01ea25ec9666d1c`
control_totals	14 rows	`cc9e55ee18dc01c4`
gl_balances	6 rows	`35f3ddd2b64253ad`
fx_rates	27 rows	`cba0676dbf3ce091`

Each input is fingerprinted with SHA-256 on ingest and recorded in the run audit trail.

3. Data quality

Rules run on the raw files. Each belongs to one of six dimensions: completeness, validity, uniqueness, consistency, integrity, timeliness. A rule is a declaration (data set, column, check, severity), so that it can be read and argued with, and it is applied in Python and exported as SQL, the two being tested against each other. Each rule has a required pass rate by severity, agreed at kickoff: below it the rule fails, and within a small gap it warns. The score of a data set or of the whole is the pass rate weighted by severity (critical 3, major 2, minor 1), from 0 to 100. IBANs are tested with the ISO 13616 check digits (remainder 1 on division by 97). Failure rates are also shown by month and source system, to see when quality changed and where.

4. Completeness

Each extract is compared, month by month, with the control total of its source: rows and sum. The bank file is proved a second way: opening balance + the lines of the day = closing balance per the bank, every day. A day that does not roll has lines missing from the file.

5. Customer master

Duplicate candidates are found by a shared tax id or IBAN, or by similar names: names are normalised (case, punctuation, legal forms), compared within blocks, must carry the same numbers and must reach an agreed similarity. Pairs are candidates for the data owner.

6. Cleaning for the reconciliation

A copy is cleaned; the rules have already measured the original. Dates are read in both formats, references put in one form and read from the bank narrative, exact repeated rows removed, a blank invoice currency taken from the customer master. A row with an unknown account, an unreadable date or no amount is excluded and listed. Ledger entries with their own id are never removed: a double posting is a break the matching has to see.

7. Matching

Four passes run from the strictest to the loosest, each on what the earlier ones left: (1) same account, reference and amount, dates close; (2) same account and amount, dates close, the references not disagreeing; (3) a bank credit against the entries of its processor batch; (4) same account and reference with a difference inside the tolerance, or a reference lost on one side with a difference inside the tolerance and dates very close. Each pair records its pass and the days between the two records.

8. Diagnosis of what is left

A reference pair with a larger difference is **transposed digits** if the digits are the same and the difference in cents divides by 9, a **bank fee** if a receipt is short by no more than the agreed fee, a **reversed sign** if the amounts are opposite, and otherwise an **amount difference**. A batch short by no more than an agreed share is an **unbooked processor fee**. The same reference and amount on another account is a **wrong account**. A second copy of a matched entry is a **duplicate posting**. A ledger entry alone is a **bank feed gap** if the balance proof fails on its day, a **stale payment** if it is an old payment, and otherwise **not at the bank**. A bank line alone is an **unrecorded charge or interest** by its narrative, a **ledger extract gap** in a month short of its control total, **not in the ledger** if it carries a reference or is a payment, and otherwise an **unidentified item**. A pair that spans the period end is **timing**.

9. Reconciliation statement

For each account, in its currency: books + what only the bank has - what only the books have + differences in matched pairs + opening difference = expected bank balance. Unexplained = balance per the bank - expected bank balance. The statement is an identity: anything left is something the data does not contain. An account with an unexplained difference should not be signed.

10. Items, age and priority

Each item carries its class, nature (timing, unrecorded, error, unknown, data gap), age at the period end, value in the reporting currency, owner and next step. High: above the agreed value, older than 90 days, or a data gap. Timing items are low.

11. Tie-out

The open invoices at each month end, in the reporting currency, against the control account. Invoices above the plausible maximum of rule I-05 are left out and listed, because they are data errors.

12. Effort

Minutes to clear an item, by nature, agreed with the team, summed per month and scaled to a year.

13. Assessment

Twenty-two tests in five areas (completeness, data quality, reconciliation, breaks, effort) give PASS, WARN or FAIL against thresholds agreed at kickoff; the overall view is Reliable (all pass), Reliable with actions (some warn) or Not reliable (some fail), with the action each non-passing test implies.

14. Commentary

Sentences are assembled by rules from the tables. The analyst reads the draft and may add notes.

15. Reproducibility

Inputs, configuration and code version are recorded in `audit.json`. Every table in the deliverables is a CSV in the run folder.

16. Limitations

The tests above cannot detect an error that is in the source system and in its control total alike. A pair is a strong indication, and a reviewer can overturn it. The class of an item is an inference. A rule catches only the defect it describes. Clearance can be seen only up to the last bank date received. No deviations from the standard methodology.

Appendix A. Parameters of this run

Parameter	Value
Reference window	25 days
Date window	9 days
Tolerance	1.0 in account currency
Bank fee limit	30.0
Batch fee share	2.0%
Stale after	60 days
Required pass rate	critical 99.9%, major 99.0%, minor 95.0%
High priority from	5,000 USD

Appendix B. Worked examples (made-up figures)

IBAN check. For GB82 WEST 1234 5698 7654 32: move the first four characters to the end, turn letters into numbers (A = 10 ... Z = 35), and divide the resulting integer by 97. A valid IBAN leaves 1.

Transposed digits. Booked 1,523.40, received 1,532.40. The difference is 9.00, which is 900 cents and divides by 9, and both amounts use the same digits. The class is transposed digits.

Statement. Books 1,000; a payment of 200 only in the books (-200 would have left the bank, so the bank is 200 higher: +200); a charge of 15 only at the bank (-15). Expected bank = 1,000 + 200 - 15 = 1,185. The bank says 1,185. Unexplained = 0.

Balance proof. Opening 500, lines of the day +120, closing per the bank 700. Unexplained = 700 - 500 - 120 = 80: lines worth 80 exist that the file does not hold.

Appendix C. Rule catalogue

Rule	Data set	Dimension	Column	Check	Severity
C-01	customers	Uniqueness	customer_id	unique	critical
C-02	customers	Completeness	name	not_null	major
C-03	customers	Completeness	tax_id	not_null	major
C-04	customers	Validity	iban	iban	critical
C-05	customers	Validity	country	in_set	major
C-06	customers	Validity	status	in_set	minor
I-01	invoices	Uniqueness	invoice_id	unique	critical
I-02	invoices	Integrity	customer_id	ref	critical
I-03	invoices	Integrity	customer_id	ref	minor
I-04	invoices	Validity	issue_date	regex	major
I-05	invoices	Validity	amount	range	critical
I-06	invoices	Completeness	currency	not_null	major
I-07	invoices	Consistency	due_date	compare	major
I-08	invoices	Consistency	paid_date	required_if	major
L-01	ledger	Uniqueness	entry_id	unique	critical
L-02	ledger	Completeness	reference	not_null	major
L-03	ledger	Validity	reference	regex	minor
L-04	ledger	Integrity	account	ref	critical
L-05	ledger	Validity	value_date	regex	major
B-01	bank	Uniqueness	line_id	unique	critical
B-02	bank	Completeness	narrative	not_null	minor
B-03	bank	Integrity	account	ref	critical
T-01	control_totals	Timeliness	due_date	compare	major